

MD RAKIBUL ISLAM RAIHAN

AI Engineer — LLM Infrastructure & Multimodal Systems (2027 New Grad)

Xi'an, China | Open to Relocation | Available July 2027

Nationality: Bangladesh | Sex: Male | DOB: 2002.12.06

Languages: Bengali (Native) | English (Fluent) | Chinese (Conversational)

Work Authorization: Requires China work visa (employer-sponsored Z Visa) | Global: Sponsorship welcome

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🌐 GitHub: github.com/theraihanrakibb 🔗 LinkedIn: [linkedin.com/in/theraihanrakib](https://www.linkedin.com/in/theraihanrakib) 🌐 Portfolio: theraihanrakibb.github.io/Online-Portfolio



SUMMARY

2027 New Grad AI Engineer targeting **AI Infrastructure / LLM Inference** and **Multimodal Generative AI**. At **InfiX.ai** built production LLM serving on **4x A800 / H800** (SGLang, 3-tier KV-cache offload, PD-disaggregation over Mooncake RDMA) — **92.27% cache hit, 8.3x faster TTFT, 192→256 concurrency**. Research: **first author** of IEEE TNNLS (Q1) **NVFP4-DiT**, a 4-bit audio-guided video diffusion transformer (**4x memory reduction**, A800/H100). Also built **multimodal audio-visual deepfake detection**. Strong in LLM inference/serving, GPU & low-precision (FP8/FP4) optimization, and multimodal diffusion/generative modeling. Python/C++/CUDA; PyTorch, Hugging Face, SGLang, Linux.

EDUCATION

M.Eng. in Software Engineering, School of Software, Northwestern Polytechnical University (985/211) - Sep 2024 - Jul 2027

GPA: 88/100 | Top 1% | Chinese Government Scholarship | President of International Student Union (ISU)

B.Eng. in Computer Science & Technology, School of Computer Science, Northwestern Polytechnical University (985/211) - Sep 2020 - Jul 2024

GPA: 85/100 | Top 1% | Presidential Scholarship | Best Undergraduate Thesis Award

EXPERIENCE

AI Infrastructure Engineer Intern | InfiX.ai | Shenzhen, China | Apr 2026 - Jun 2026

- Cut TTFT 8.3x (55s → 6.7s) by building a 3-tier KV-cache offload (GPU HBM → CPU DRAM → SSD) over Mooncake RDMA on 4x A800, reaching 92.27% cache-hit for long-context models.
- Patched SGLang source across cache-management and memory-allocation modules (kvcacheio.py, hiradix_cache.py, memory_pool_host.py) to fix production cache failures and deploy hybrid-attention Qwen models.
- Built PD-disaggregated (prefill/decode) serving with RDMA KV sharing, raising stable concurrency 192 → 256 req and boosting GPU utilization via RadixAttention prefix-caching and continuous batching.
- Profiled CUDA/Triton inference with Nsight Systems, localized GEMM bottlenecks, and improved serving throughput via kernel optimization and FP8 quantization.
- Applied Little's Law to model capacity limits, proving long-context serving on 4x A800-80GB was bounded to ~1.5 RPS and guiding scale-out.
- Diagnosed and resolved 12+ production incidents across RDMA, etcd, networking, storage, and distributed inference.
- Deployed Qwen models on A800 clusters (TP2 single-node, TP4 with PD-disaggregation) and ran DeepSeek on H800 multi-node clusters (8 GPUs each), tuning tensor-parallel layouts for throughput and stability.

Software Engineer Intern (AI Agent) | Hong Kong Amram Group | Shenzhen, China | Jun 2025 - Sep 2025

- Developed production full-stack applications using React, REST APIs, and backend automation services.
- Built AI-agent workflow automation systems that improved operational efficiency by **25%**.

Electrical Software Engineer Intern | Shaanxi Longong Intelligent Technology | Xi'an, China | Feb 2025 - Apr 2025

- Developed PLC-based industrial control systems using CODESYS for robotics and industrial automation.
- Implemented Modbus, CAN, and Ethernet/IP communication protocols, improving motor-control performance by **15%**.

RESEARCH

Graduate Researcher | Northwestern Polytechnical University | Sep 2024 - Present

Advisor: Prof. Zhang Peng

- Developed **NVFP4-DiT**, a **4-bit audio-guided video diffusion transformer** achieving **4x memory reduction** while maintaining generation quality.
- Implemented low-precision quantization (**FP8/FP4/NVFP4**) for multimodal diffusion transformers; evaluated quality-efficiency trade-offs.
- Benchmarked NVFP4-DiT on **A800/H100 GPUs**; profiled end-to-end latency and memory with PyTorch Profiler and Nsight to guide low-precision kernel choices.
- Research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.
- M.Eng. Thesis: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies."

Undergraduate Researcher | Northwestern Polytechnical University | Jan 2023 - Jul 2024

Advisor: Dr. Yaxing Chen

- Designed and built a distributed confidential-query framework on Apache Spark + CKKS homomorphic encryption, achieving **95% privacy preservation** and **20% performance improvement** via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

PROJECTS

AI Infra Portfolio | github.com/theraihanrakibb/ai-infra-portfolio

- 10 production-grade, Dockerized AI-infra projects: LLM gateway, model serving, GPU observability, training ops, MLOps registry, KV-cache simulator, benchmarking, quantization, tracing, RAG.

Distributed Training Launcher | github.com/theraihanrakibb/train-launcher

- CLI for multi-node PyTorch training with fault tolerance and automatic retry; cut manual ops overhead for distributed jobs.

C Compiler (Frontend) | github.com/theraihanrakibb/c-compiler-frontend

- Self-built C compiler (lexer/parser/codegen) using Yacc/Lex; applied systems and algorithms fundamentals.

NVFP4-DiT (Multimodal Generation) | github.com/theraihanrakibb/NVFP4-DiT

- First-author IEEE TNNLS (Q1, under review) 4-bit audio-guided video diffusion transformer; 4x memory reduction while preserving generation quality on A800/H100.

Multimodal Deepfake Detection | github.com/theraihanrakibb/M.Eng-Thesis-Multimodal-Deepfake-Audio-Visual-Temporal-Framework

- M.Eng. thesis: audio-visual-temporal framework detecting deepfake video via visual, audio, and temporal inconsistency analysis.

TECHNICAL SKILLS

AI Infrastructure & LLM Serving: SGLang (RadixAttention, continuous batching, PD-disaggregation) | vLLM | TensorRT-LLM | KV-cache design & offload | Mooncake RDMA | NCCL | etcd | NATS | prefix caching

GPU & Low-Precision Optimization: CUDA | Triton | CUTLASS | FlashAttention-2/3 | FP8 / FP4 / INT4 / NVFP4 quantization | GEMM tuning | Nsight Systems/Compute | PyTorch Profiler

Multimodal & Generative AI: PyTorch | Hugging Face (Transformers, Diffusers) | Diffusion (DDPM, DiT, audio-guided video generation) | AIGC | multimodal audio-visual understanding & fusion | deepfake detection | OpenCV

Distributed Training & MLOps: PyTorch distributed (DDP/FSDP) | LoRA/QLoRA | DeepSpeed (ZeRO) | Apache Spark | CKKS homomorphic encryption | Docker | Kubernetes | Prometheus/Grafana | CI/CD | AWS / Aliyun | Slurm

Languages & Fundamentals: Python (primary) | C++ | CUDA C++ | Java | SQL | Bash | data structures & algorithms | OS | computer networks

PUBLICATIONS & HONORS

Publication: First Author, "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" - IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), Under Review.

Honors: Chinese Government Scholarship (2024-2027); Presidential Scholarship (2020-2024); Best Undergraduate Thesis Award (2024); Outstanding Graduate (2024), NWPU; Hackathon Winner - NWPU AI Challenge (2023).

MD RAKIBUL ISLAM RAIHAN

AI 工程师 — 大模型基础设施与多模态系统 (2027 届应属)

现居: 中国·西安 | 可接受异地 | 2027 年 7 月到岗

国籍: 孟加拉国 | 性别: 男 | 出生日期: 2002.12.06

语言: 孟加拉语 (母语) | 英语 (流利) | 中文 (日常交流)

工作许可: 需办理中国工作签证 (雇主担保 Z 签证) | 全球: 接受担保

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个人简历

2027 届应属 AI 工程师, 专注大模型基础设施 / 推理与多模态生成式 AI。在 InfiX.ai 基于 4x A800 / H800 构建生产级大模型服务 (SGLang、三级 KV-cache 卸载、基于 Mooncake RDMA 的 PD 分离) —— 缓存命中率 92.27%、TTFT 降低 8.3 倍、并发 192→256。研究方向: IEEE TNNLS (Q1) 论文第一作者 NVFP4-DiT, 4 比特音频引导视频扩散 Transformer (显存降 4 倍, A800/H100); 并构建多模态音视频深度伪造检测。擅长大模型推理与服务、GPU 与低精度 (FP8/FP4) 优化、多模态扩散与生成建模。Python/C++/CUDA; PyTorch、Hugging Face、SGLang、Linux。

教育背景

软件工程硕士, 软件学院, 西北工业大学 (985/211) - 2024年9月 - 2027年7月

GPA: 88/100 | 前 1% | 中国政府奖学金 | 国际学生联合会主席 (ISU)

计算机科学与技术学士, 计算机学院, 西北工业大学 (985/211) - 2020年9月 - 2024年7月

GPA: 85/100 | 前 1% | 校长奖学金 | 最佳本科毕业论文奖

工作经历

无界智索 (深圳) 科技 (InfiX.ai) | AI 基础设施工程师实习 | 深圳 | 2026年4月 - 2026年6月

- 通过 Mooncake RDMA 在 4x A800 上构建三级 KV-cache 卸载 (GPU HBM -> CPU DRAM -> SSD), 将 TTFT 降低 8.3 倍 (55s -> 6.7s), 长上下文模型缓存命中率达 92.27%。
- 修改 SGLang 源码的缓存管理与内存分配模块 (kvcacheio.py、hiradix_cache.py、memory_pool_host.py), 修复生产缓存故障并部署混合注意力 Qwen 模型。
- 构建 PD 分离 (prefill/decode) 推理架构, 基于 RDMA 共享 KV, 稳定并发由 192 -> 256 提升, 并通过 RadixAttention 前缀缓存与连续批处理提高 GPU 利用率。
- 使用 Nsight Systems 分析 CUDA/Triton 推理链路, 定位 GEMM 瓶颈, 经算子优化与 FP8 量化提升服务吞吐。
- 运用 Little's Law 建模容量上限, 论证 4x A800-80GB 上长上下文服务受限于约 1.5 RPS, 指导扩容。
- 排查并解决 12+ 起生产故障 (RDMA、etcd、网络、存储与分布式推理)。
- 在 A800 集群部署 Qwen 模型 (单节点 TP2, 以及 TP4 PD prefill/decode 分离), 并在 H800 多节点集群 (每集群 8 卡) 运行 DeepSeek, 调优张量并行布局以提升吞吐与稳定性。

香港阿姆兰集团 (Amram Group) | 软件工程师实习 (AI Agent) | 深圳 | 2025年6月 - 2025年9月

- 基于 React、REST API 与后端自动化服务开发生产级全栈应用。
- 开发 AI Agent 工作流自动化系统, 运营效率提升 25%。

陕西龙工智能科技 (Longong) | 电气软件工程师实习 | 西安 | 2025年2月 - 2025年4月

- 基于 CODESYS 开发 PLC 工业控制系统, 用于机器人与工业自动化。
- 实现 Modbus、CAN 与 Ethernet/IP 通信协议, 电机控制性能提升 15%。

研究经历

研究生研究员 | 西北工业大学 | 2024年9月 - 至今

导师: 张蓬 教授

- 开发 NVFP4-DiT: 4 比特音频引导视频扩散 Transformer, 显存降低 4 倍且保持生成质量。
- 实现多模态扩散 Transformer 的低精度量化 (FP8/FP4/NVFP4), 评估质量-效率权衡。
- 在 A800/H100 上对 NVFP4-DiT 进行基准测试, 使用 PyTorch Profiler 与 Nsight 分析端到端延迟与显存, 指导低精度算子选型。
- 基于视觉、音频与时序一致性的多模态深度伪造检测研究。
- 硕士论文: 《检测深度伪造视频: 基于多模态音视频框架与时序不一致性》。

本科研究员 | 西北工业大学 | 2023年1月 - 2024年7月

导师: 陈亚星 博士

- 基于 Apache Spark 与 CKKS 同态加密设计并实现分布式机密查询框架, 通过优化分析管线实现 95% 隐私保护与 20% 性能提升。
- 本科论文: 《面向 Apache Spark 的分布式机密查询协议设计与实现》。

项目

AI 基础设施代码集 | github.com/theraihanrakibb/ai-infra-portfolio

- 10 个生产级、Docker 化的 AI 基础设施项目: LLM 网关、模型服务、GPU 可观测性、训练运维、MLOps 注册表、KV-cache 模拟器、基准测试、量化、链路追踪、RAG。

分布式训练启动器 | github.com/theraihanrakibb/train-launcher

- 用于启动与监控多节点 PyTorch 训练任务的命令行工具, 具备容错与自动重试; 降低分布式训练运维成本。

C 编译器 (前端) | github.com/theraihanrakibb/c-compiler-frontend

- 基于 Yacc/Lex 的自研 C 编译器前端 (词法/语法分析/代码生成); 体现系统与算法基础。

NVFP4-DiT (多模态生成) | github.com/theraihanrakibb/NVFP4-DiT

- 第一作者 IEEE TNNLS (Q1, 审稿中) 4 比特音频引导视频扩散 Transformer; 在 A800/H100 上显存降低 4 倍且保持生成质量。

多模态深度伪造检测 | github.com/theraihanrakibb/M.Eng-Thesis-Multimodal-Deepfake-Audio-Visual-Temporal-Framework

- 硕士论文: 基于视觉、音频与时序不一致性分析的多模态深度伪造视频检测框架。

技术技能

大模型基础设施与服务: SGLang (RadixAttention、连续批处理、PD 分离)、vLLM、TensorRT-LLM、KV Cache 设计与卸载、Mooncake RDMA、NCCL、etcd、NATS、前缀缓存

GPU 与低精度优化: CUDA、CUTLASS、Triton、FlashAttention-2/3、FP8 / FP4 / INT4 / NVFP4 量化、GEMM 调优、Nsight Systems/Compute、PyTorch Profiler

多模态与生成式 AI: PyTorch、Hugging Face (Transformers、Diffusers)、扩散 (DDPM、DiT、音频引导视频生成)、AIGC、多模态音视频理解与融合、深度伪造检测、OpenCV

分布式训练与 MLOps: PyTorch 分布式 (DDP/FSDP)、LoRA/QLoRA、DeepSpeed (ZeRO)、Apache Spark、CKKS 同态加密、Docker、Kubernetes、Prometheus/Grafana、CI/CD、AWS / 阿里云、Slurm

编程语言与基础: Python (主语言)、C++、CUDA C++、Java、SQL、Bash、数据结构与算法、操作系统、计算机网络

论文与荣誉

论文: 第一作者, 《NVFP4-DiT: 面向低成本视频生成的高效 4 比特音频引导视频扩散 Transformer》- IEEE TNNLS (Q1), 审稿中。

荣誉: 中国政府奖学金 (2024-2027); 校长奖学金 (2020-2024); 最佳本科毕业论文奖 (2024); 西北工业大学优秀毕业生 (2024); NWPU AI 挑战赛黑客松冠军 (2023)。